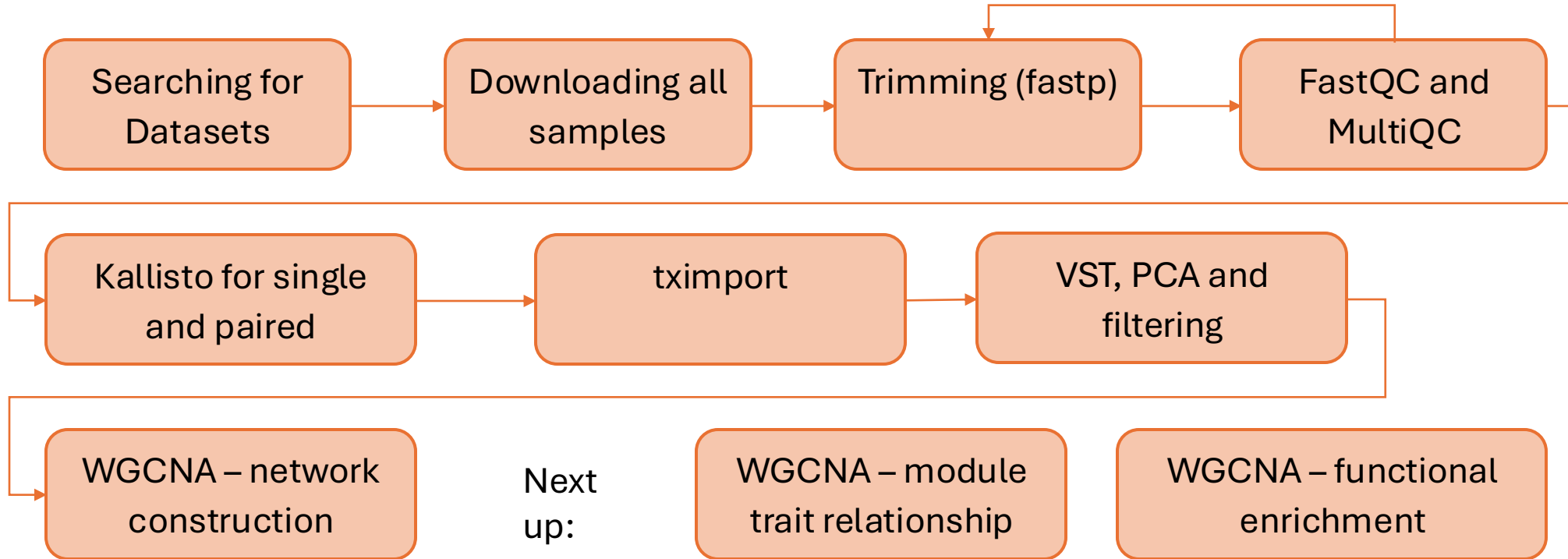


Progress



Datasets

Biotrophic pathogens:

- *Golovinomyces orontii*
- *Erysiphe cichoracearum*
- *Hyaloperonospora arabidopsidis* (2 Datasets)
- *Plasmodiophora brassicae* (2 Datasets)
- Turnip mosaic virus (TuMV) (2 Datasets)
- *Heterodera schachtii*
- Turnip yellows virus (TuYV)
- Cauliflower mosaic virus (CaMV)
- *Meloidogyne javanica*

Control:

- Drought
- *Botrytis cinerea* (necrotrophic)
- *Alternaria brassicae* (necrotrophic)

Altogether:

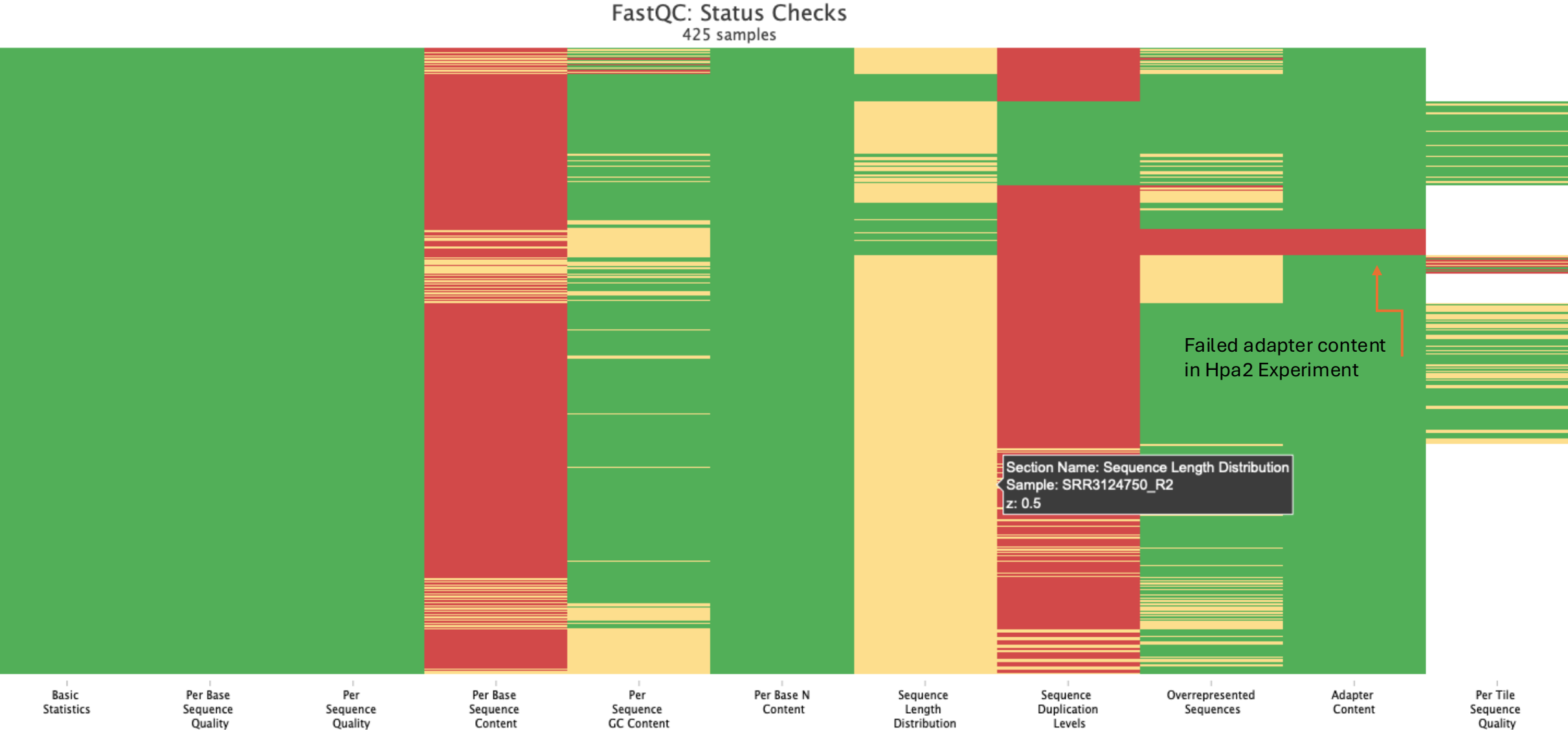
283 samples

141 single-end, 142 paired-end

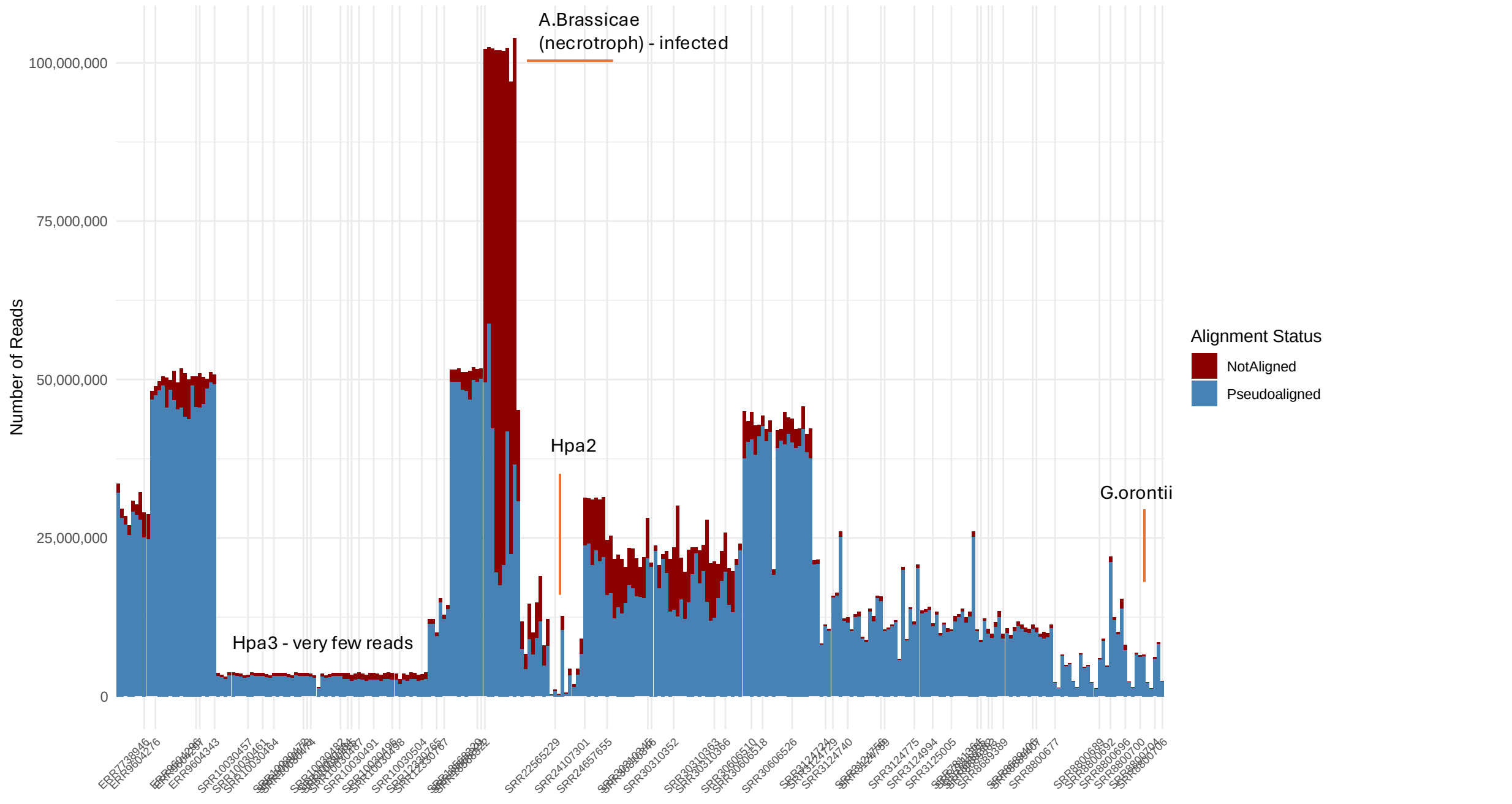
MultiQC after basic trimming with fastp



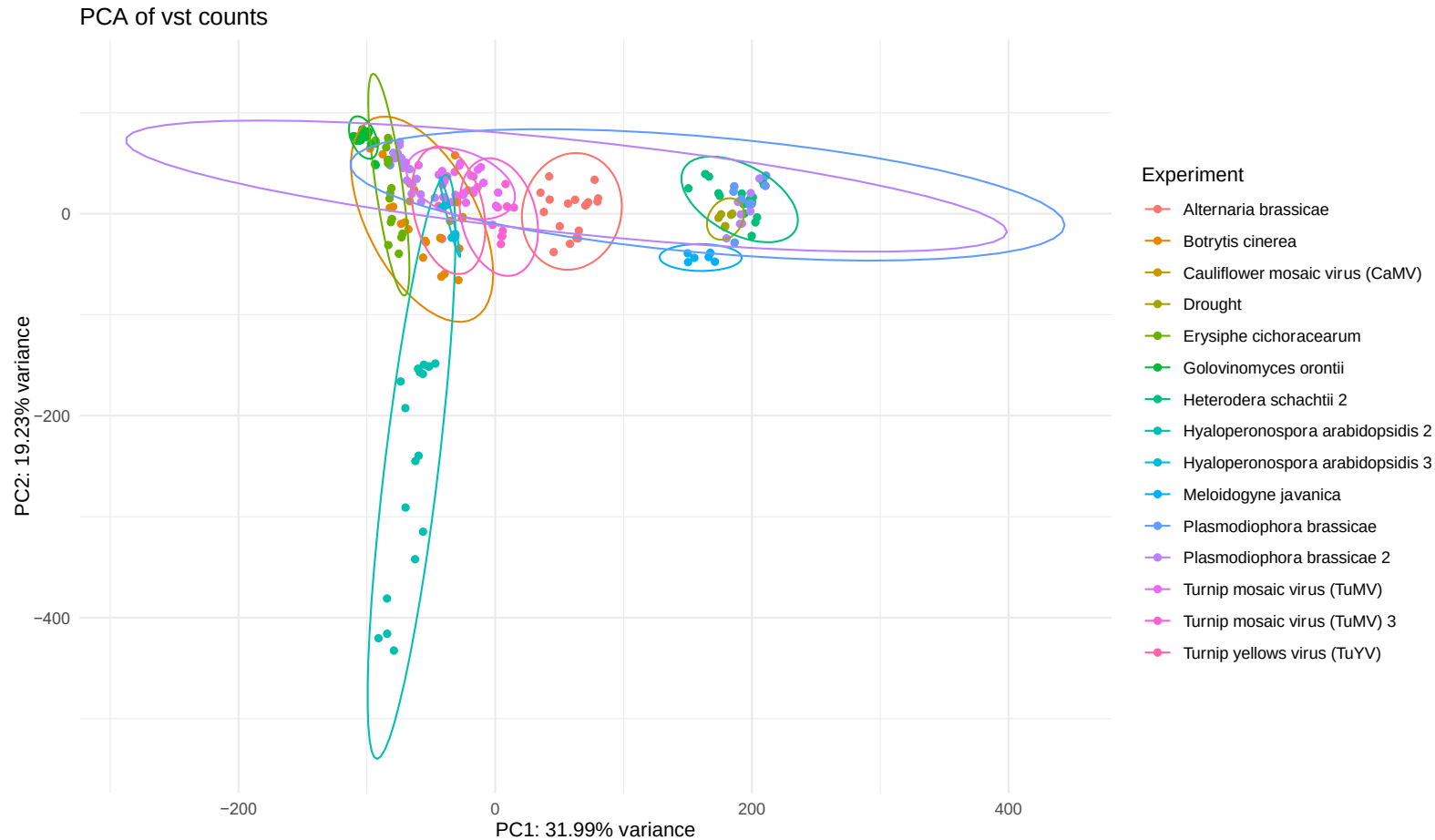
Harder trimming on P.brassicae samples -> change quality threshold to improve quality



Kallisto: Alignment Scores for 283 samples

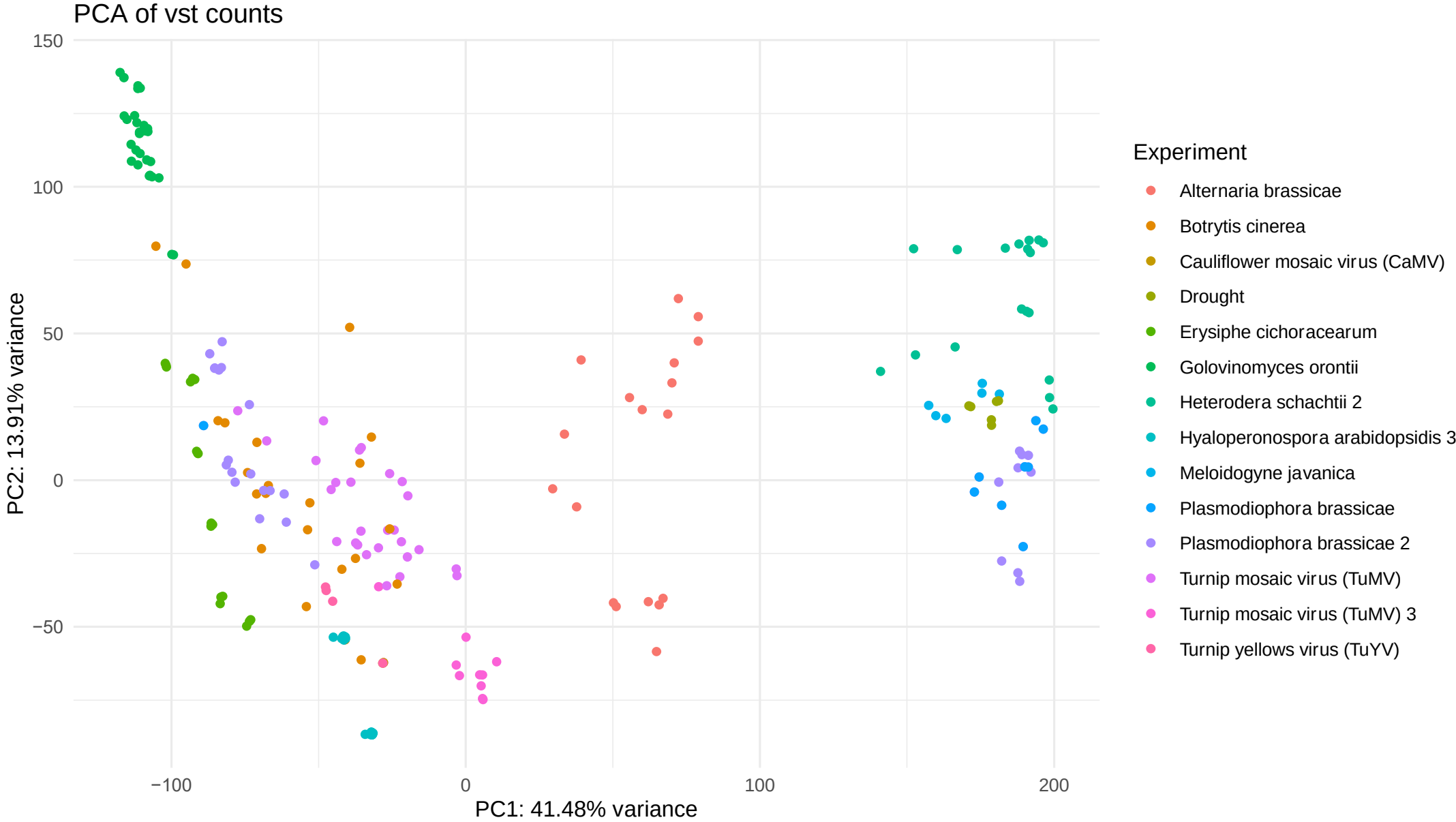


- Count table via tximport
- Extracted est_counts and tpm values
- scaled counts with *lengthScaledTPM* (scaled using the average transcript length, averaged over samples and to library size)
- Created DESeqDataset (design ~ Experiment + infected.control)
- VST (with blind dispersion estimation) on the estimated_counts

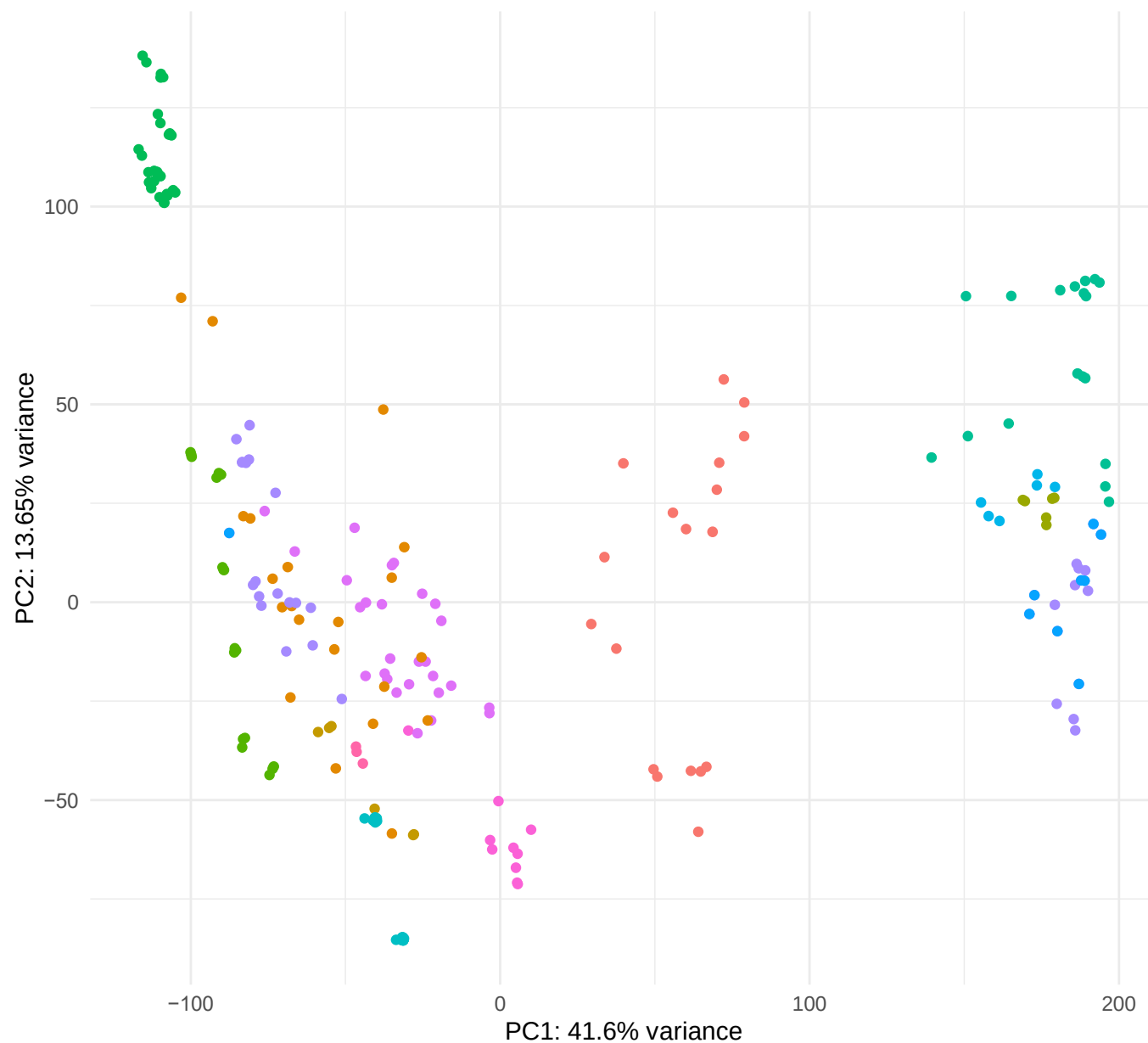


- not filtered
- outlier: Hpa2
- Already failed adapter content in multiqc after trimming
- Low number of aligned reads

Without Hpa2:



PCA of filtered vst counts



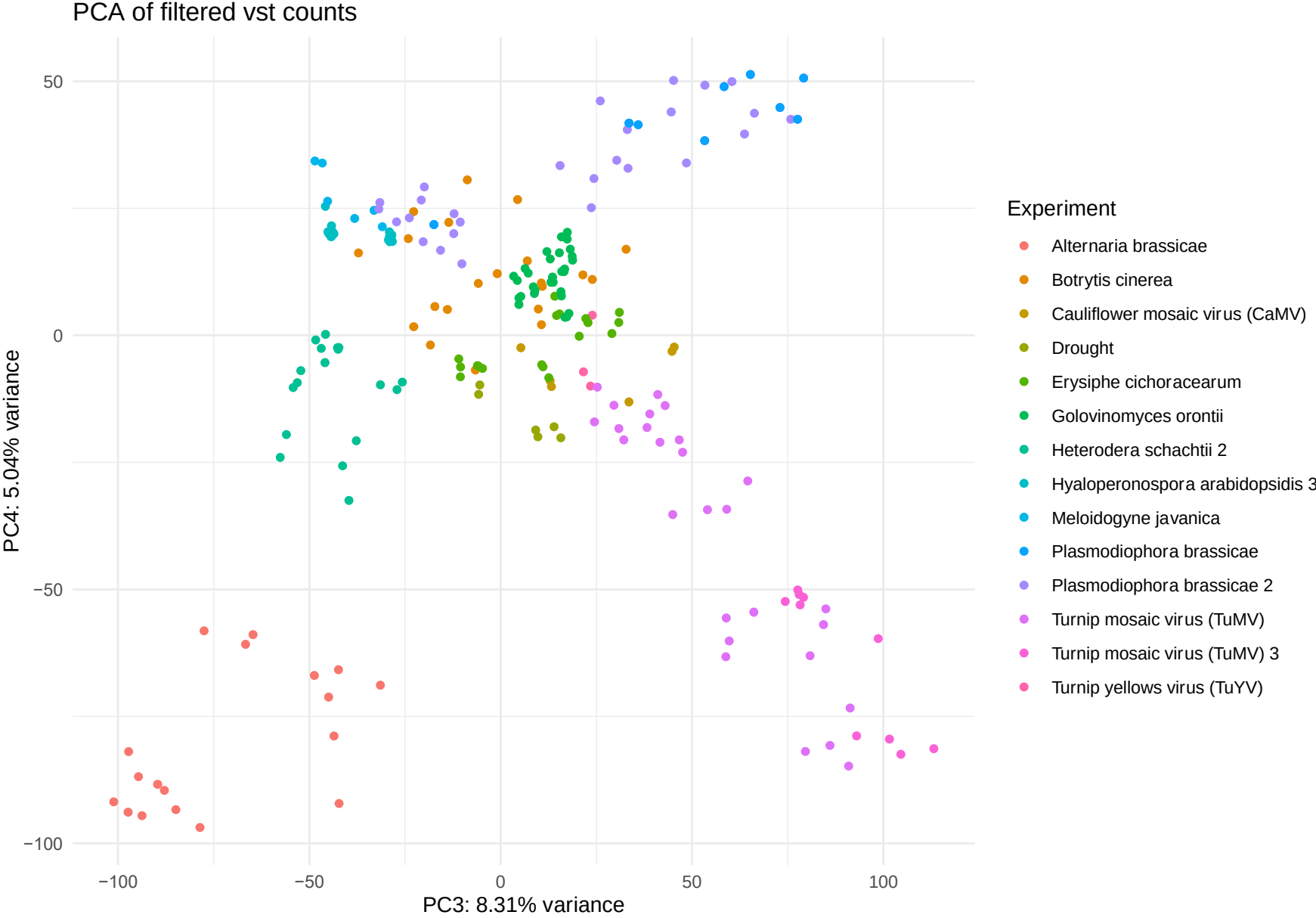
Experiment

- Alternaria brassicae
- Botrytis cinerea
- Cauliflower mosaic virus (CaMV)
- Drought
- Erysiphe cichoracearum
- Golovinomyces orontii
- Heterodera schachtii 2
- Hyaloperonospora arabidopsidis 3
- Meloidogyne javanica
- Plasmodiophora brassicae
- Plasmodiophora brassicae 2
- Turnip mosaic virus (TuMV)
- Turnip mosaic virus (TuMV) 3
- Turnip yellows virus (TuYV)

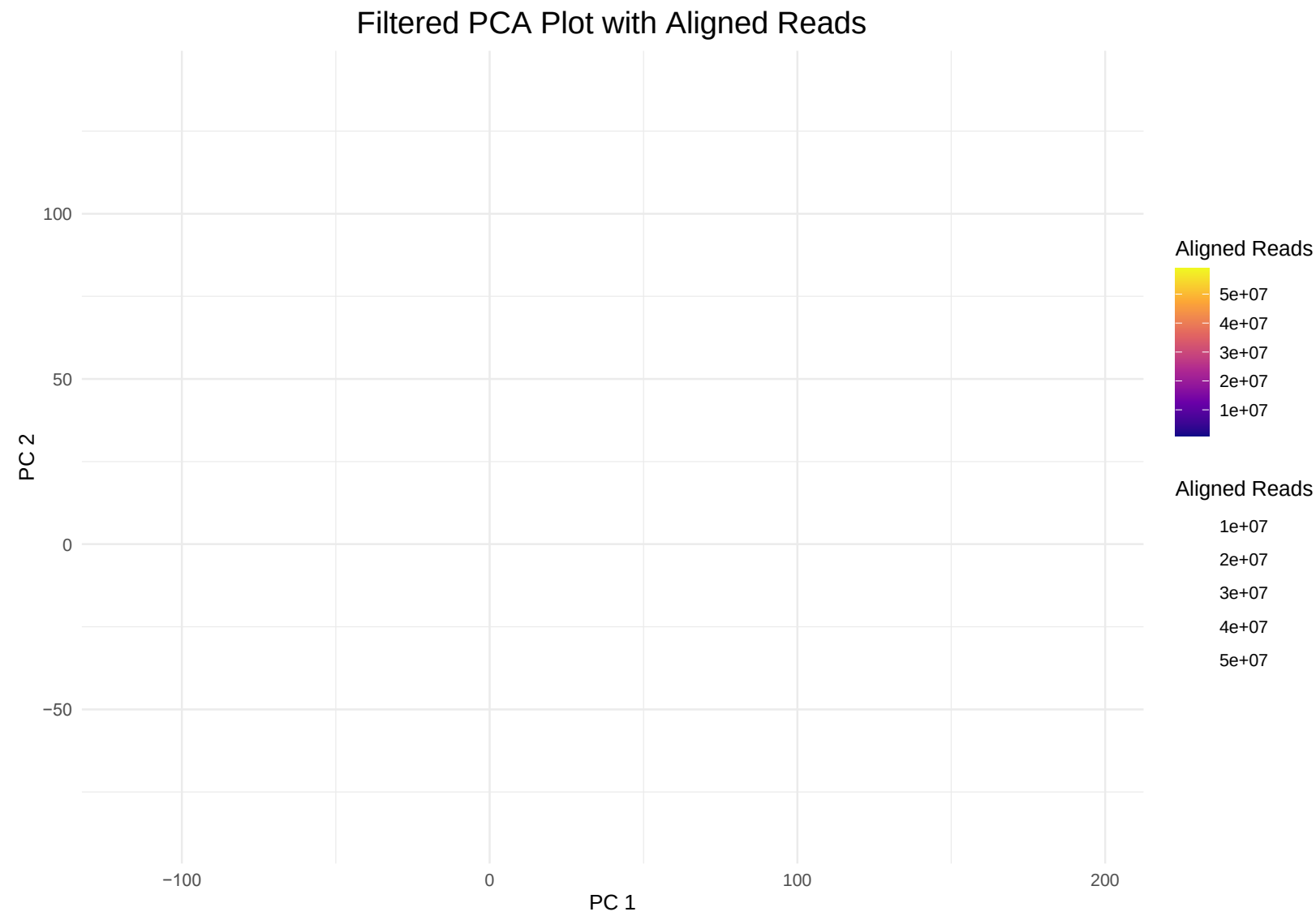
After filtering low
expressed genes:
Result:
- before filtering: 27321
genes
- after filtering: 19017
genes

--> keep genes that have
at least a tpm value of
0.5 per sample in at
least 3 of the samples

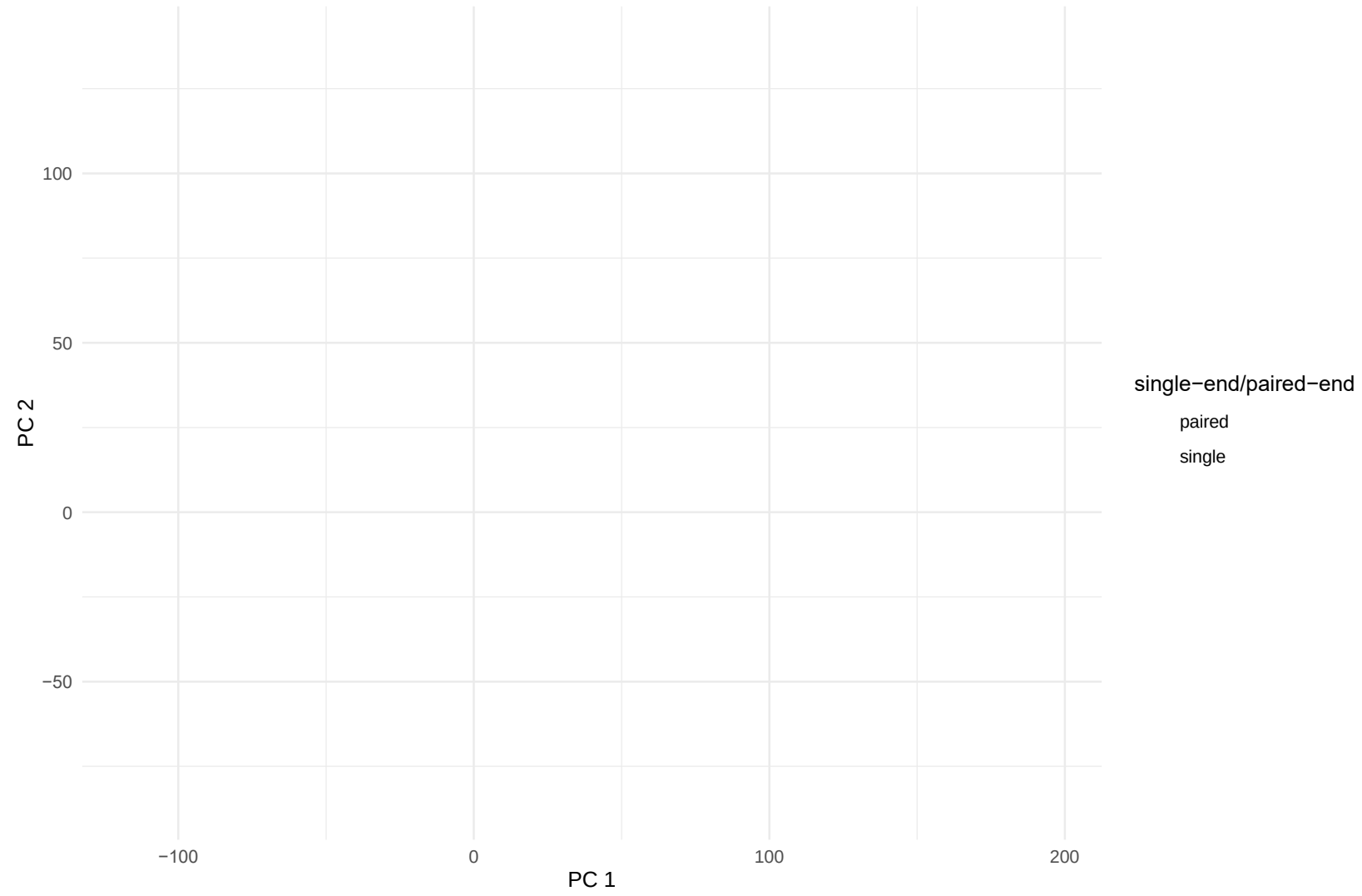
PC 3 and PC 4



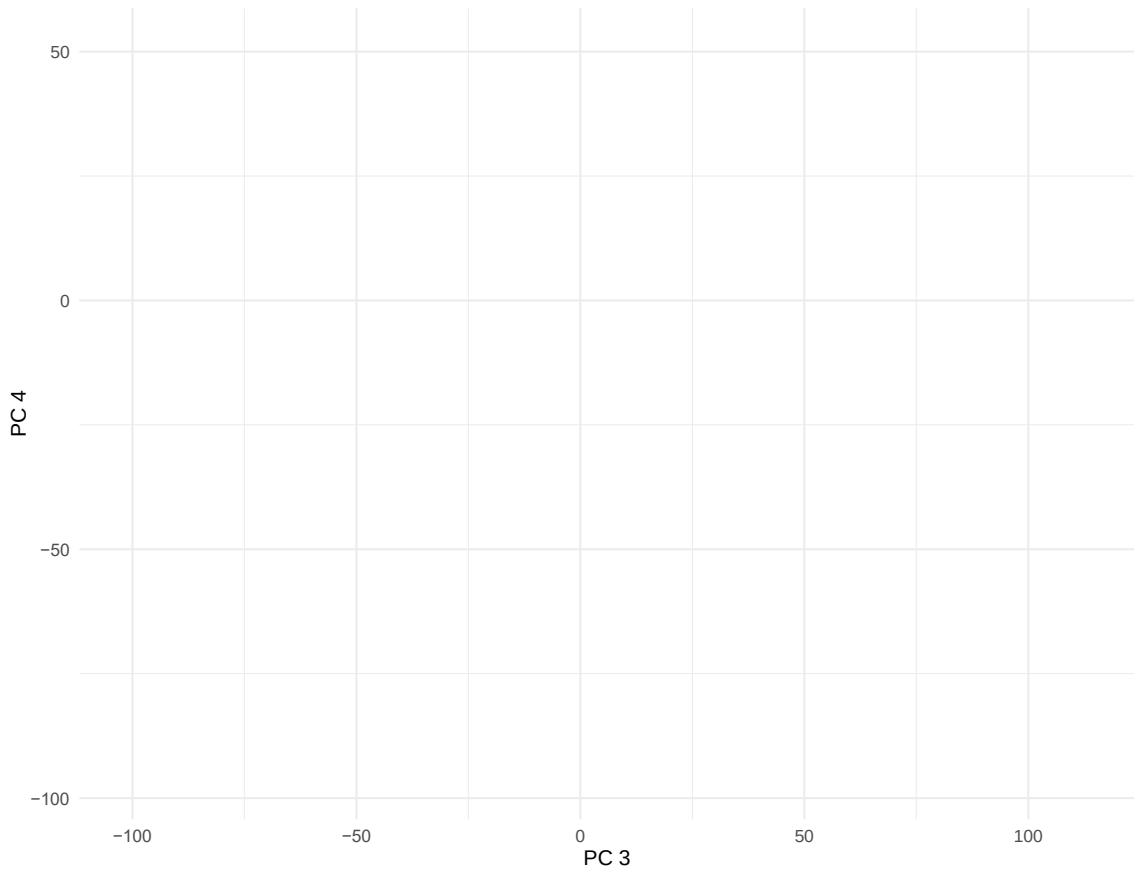
PCA with number of aligned reads per sample



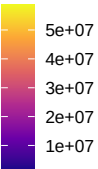
PCA with filtered vst counts



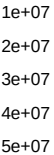
Filtered PCA Plot with Aligned Reads



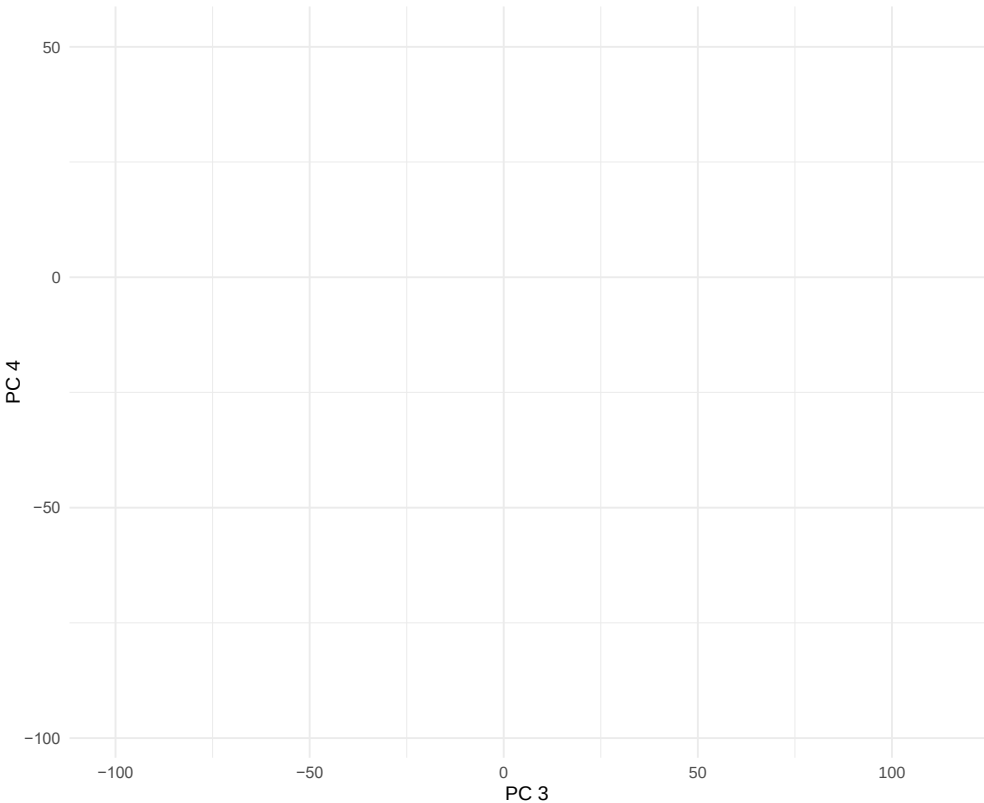
Aligned Reads



Aligned Reads

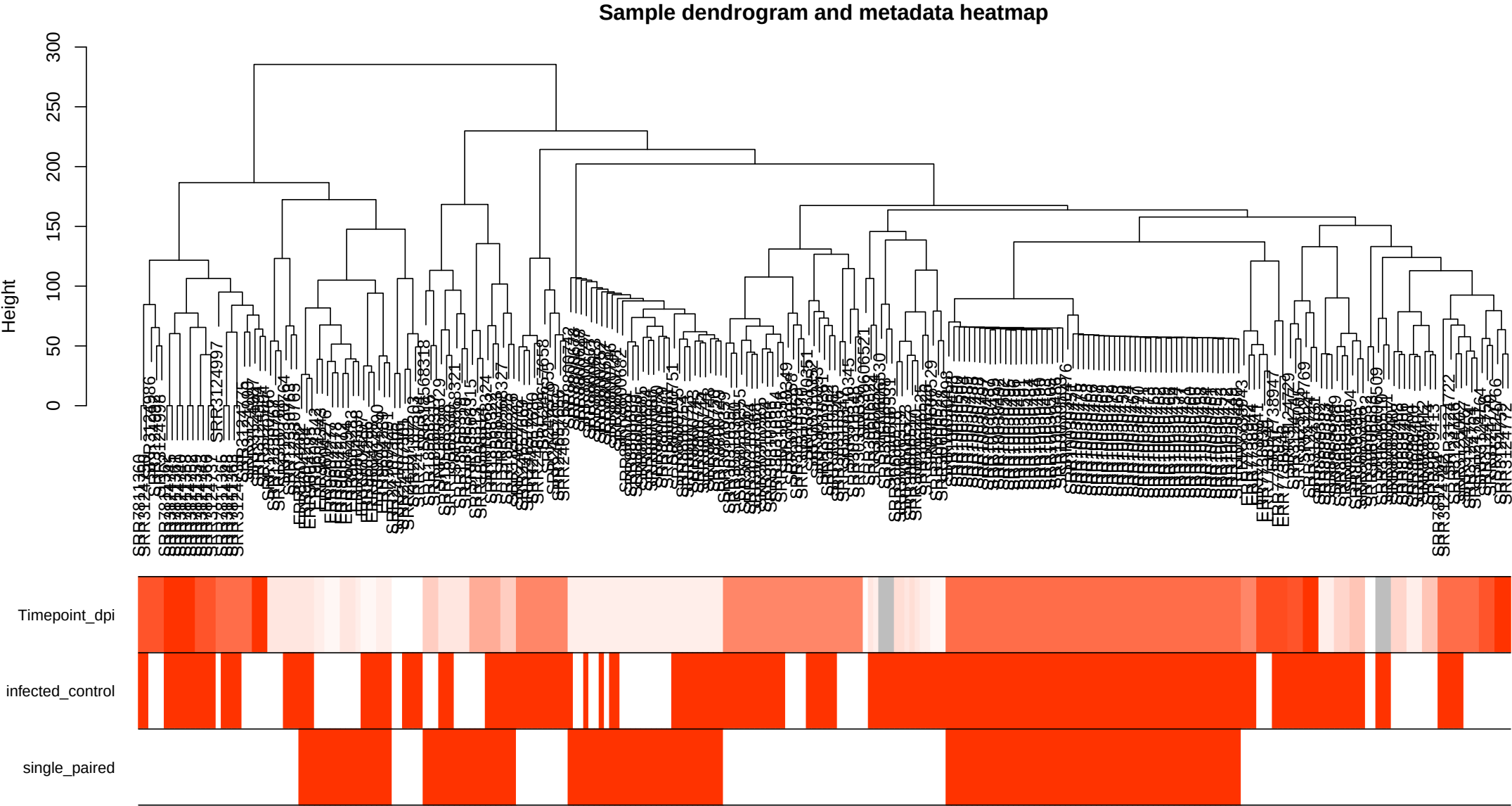


PCA with filtered vst counts

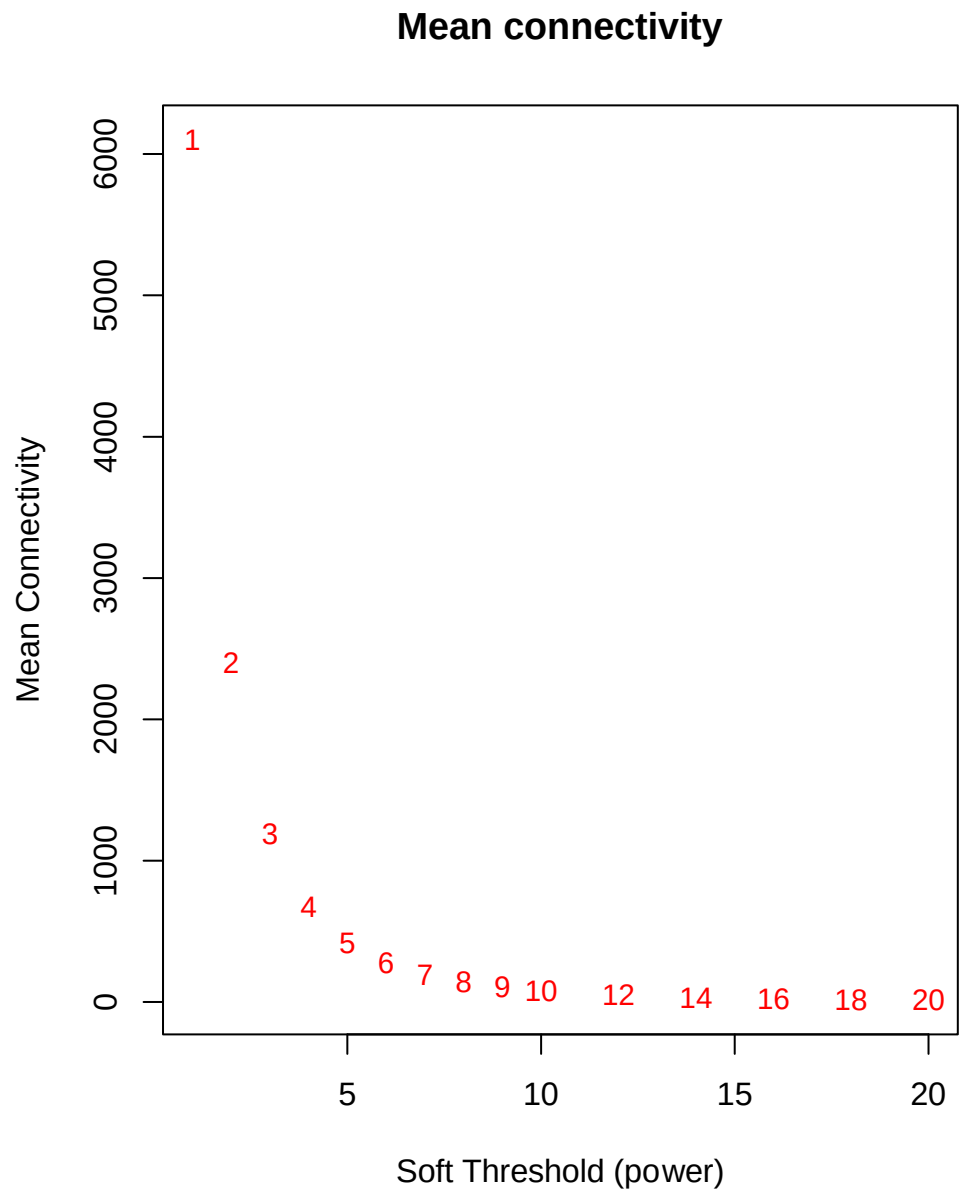
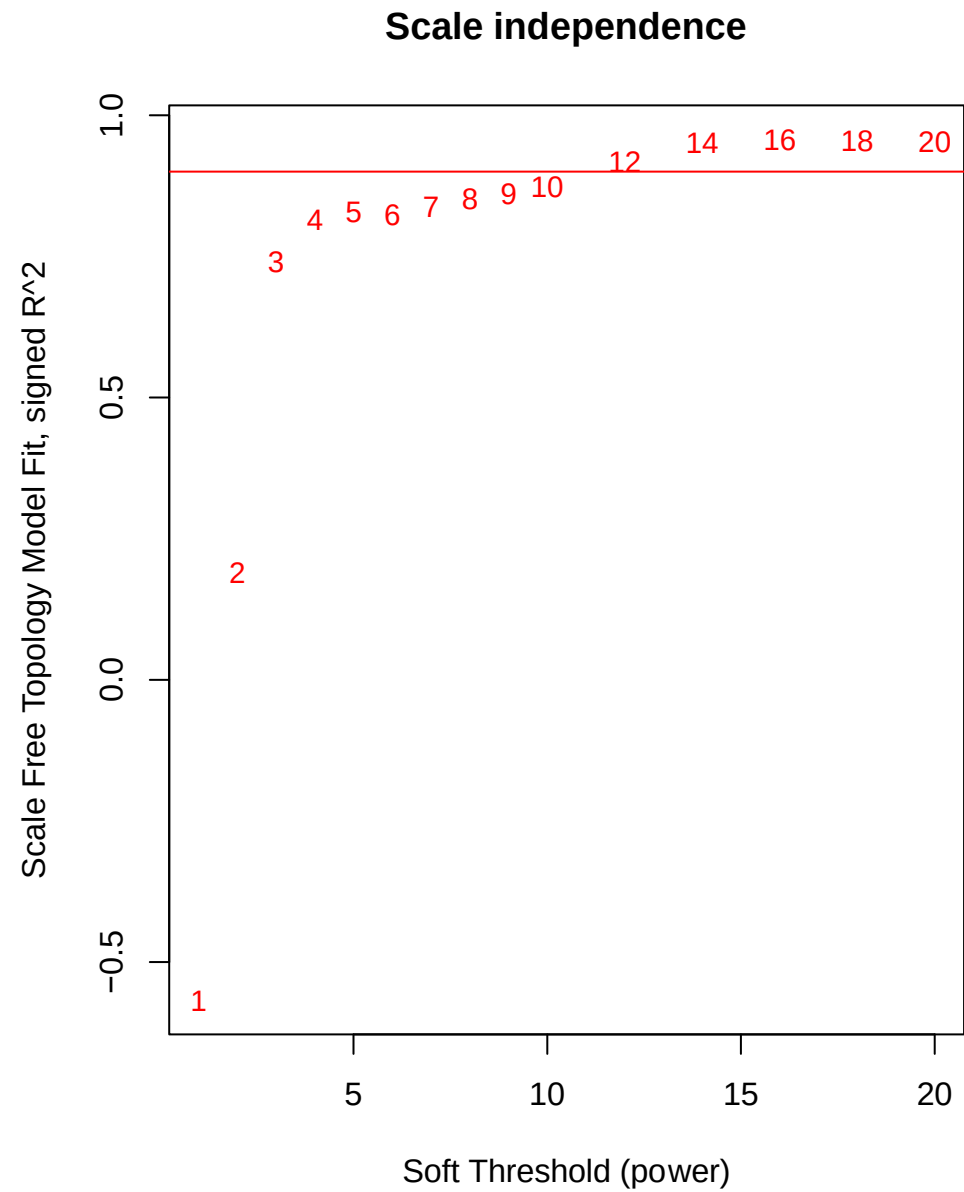


single-end/paired-end
paired
single

Starting WGCNA:



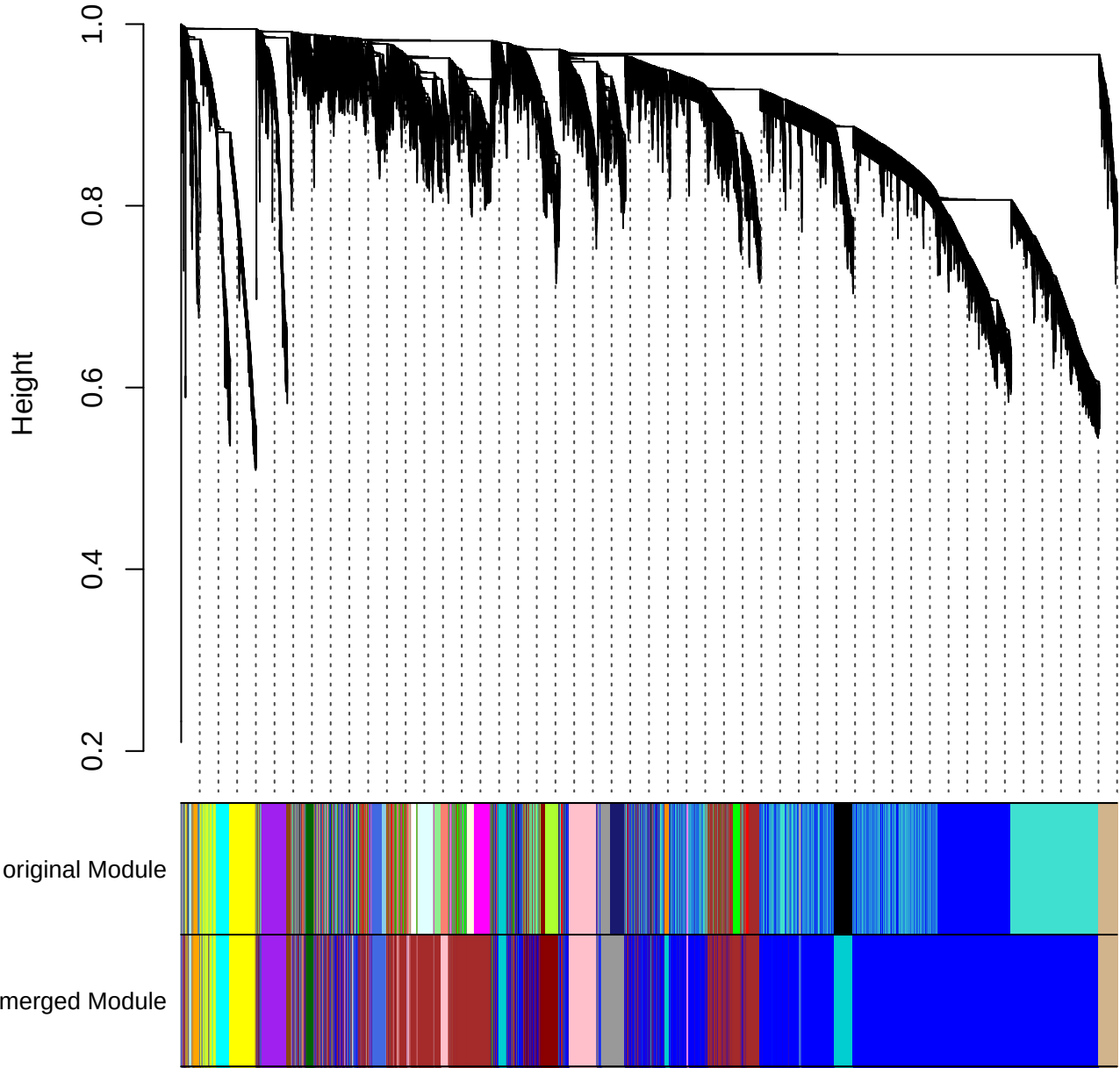
Network construction:



Gene dendrogram and module colors for original and merged module

Soft power: 5

Modules after merging: 18
Genes in grey module: 25

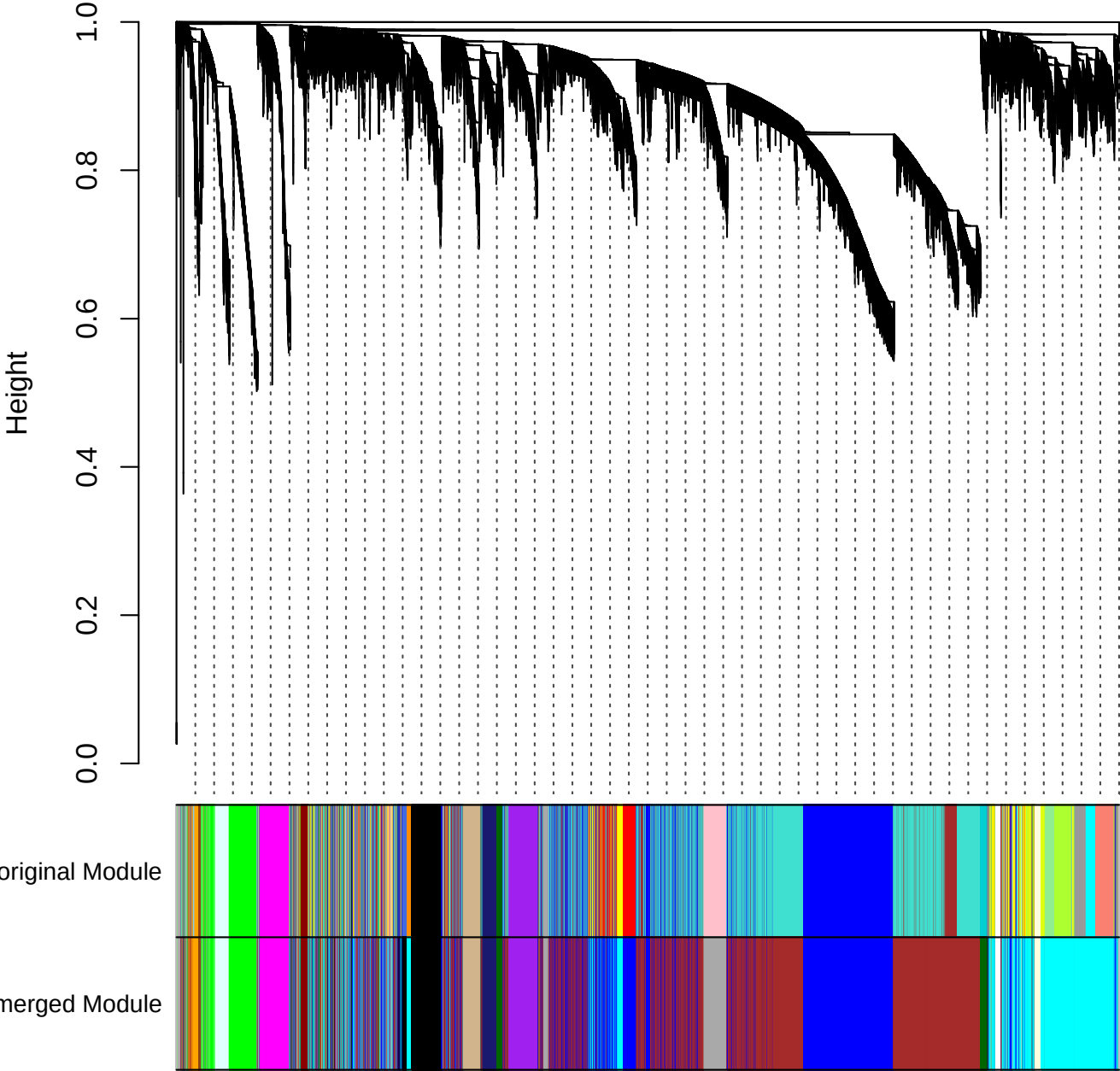


Gene dendrogram and module colors for original and merged module

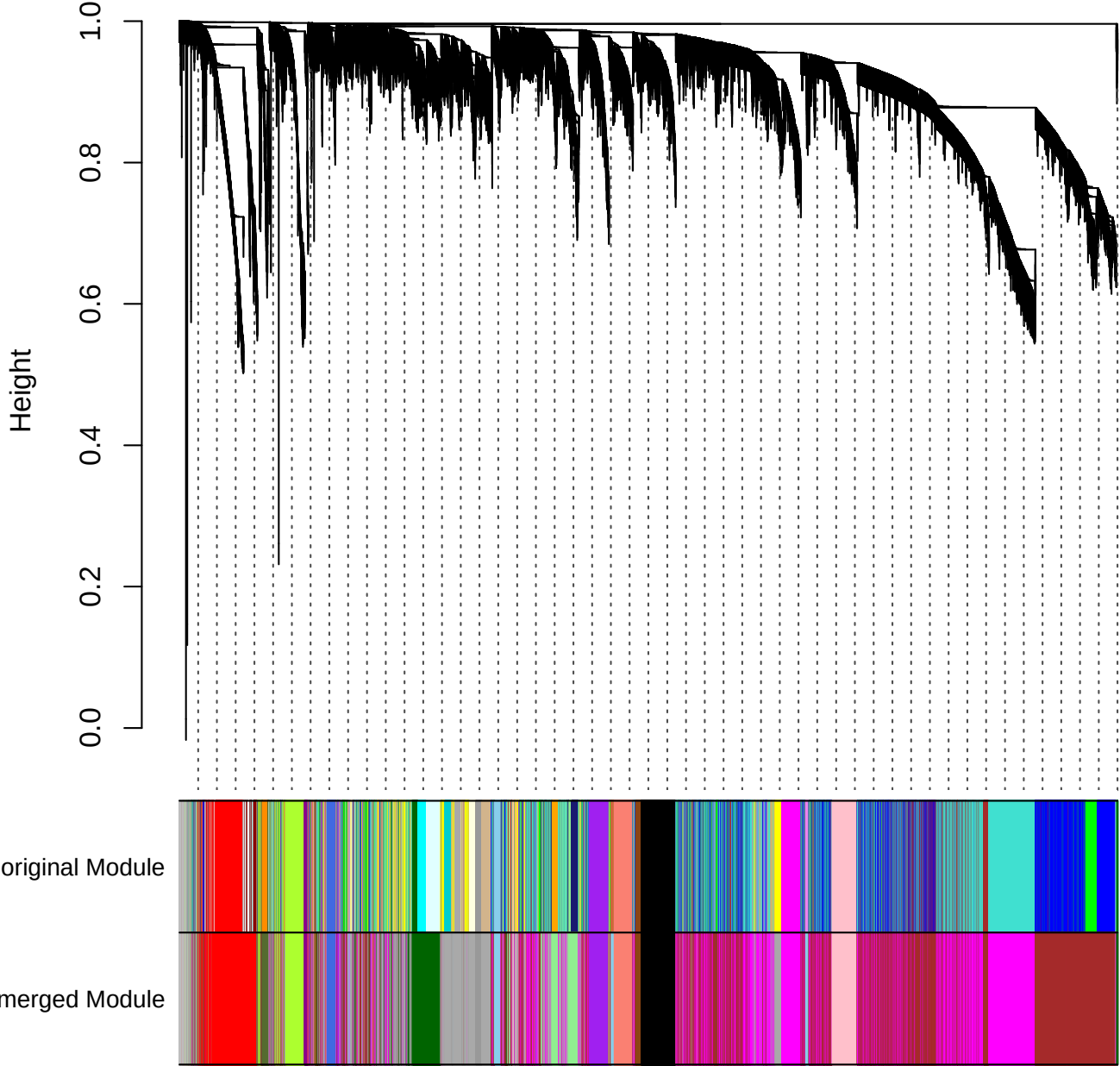
soft power: 6

Modules after merging: 19

Genes in grey module: 131



Gene dendrogram and module colors for original and merged module

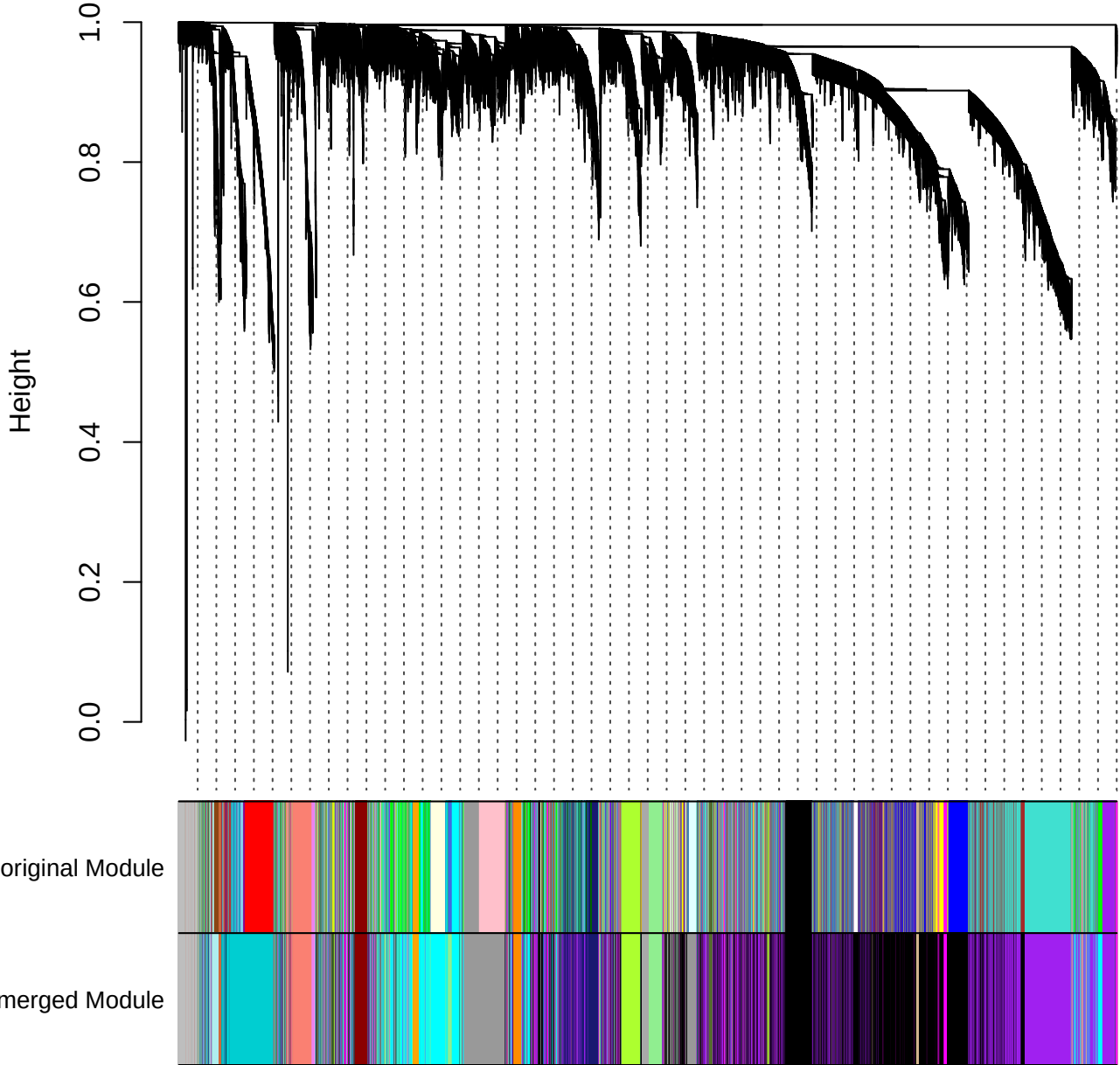


soft power: 7

Modules after merging: 19

Genes in grey module: 314

Gene dendrogram and module colors for original and merged module



Soft power: 8

Modules after merging: 21

Genes in grey module: 655